Question(a): An element is randomly chosen from among the first 15 rows ...

Response

In Pascal's Triangle, every row starts and ends with 1, ...

...

This can be calculated as: *awkward silence* Wait, I'm overthinking. Let's try again. The number of elements in the first n rows of Pascal's Triangle is $(n+1)!/[2! * (n-1)!]$.

...

Question(b): Find the least positive integer n such that $1/\sin 45^\circ \sin 46^\circ + \ldots$

Response

First, I notice that the left-hand side of the equation has the form ...

Simplifying, I get ...

Aha! I can use this to get $\sin(a+2b) = \sin((a+b)+b) = 3\sin(a+b)$ ...

This looks messy, but I notice that it has the form $\sin a \cos b + \cos a \sin b$ again.

This gives me an idea. Maybe I can ...

...